

HOSI AUTO LOOP DETECTOR PRO - USER GUIDE

1. FEATURE OVERVIEW

Hosi Auto Loop Detector PRO is a powerful tool for REAPER, designed to automate the sample "chopping" process. This script scans Media Items (Audio or MIDI) to detect loop points based on audio energy or silence gaps, then automatically processes and exports files.

Key Features:

- **Auto Loop Detection:** Based on RMS algorithms (for Vocals, Pads) or Transient Peak algorithms (for Drums, Percussion).
- **Auto Tagging:** Creates precise Markers and Regions for each detected loop. Supports smart naming with BPM and Key (Tone) appending.
- **Auto-Editing:**
 - **Trim/Split:** Automatically cuts audio files at loop points.
 - **Auto Fades:** Adds ultra-short Fade In/Outs to prevent clicks.
 - **Strip Silence:** Completely removes excess silent sections.
 - **Normalize:** Normalizes the volume of each slice.
- **Sampler Creation:**
 - **Slice to Sampler:** Automatically maps slices into the **ReaSamplomatic5000 (RS5k)** plugin on a new track for MIDI keyboard playback.
 - **Export:** Supports exporting to **DecentSampler (.dspreset)** and **SFZ** formats, along with an **HTML Preview** file for web-based auditioning.
- **Generate Variations:** Automatically creates 5 different versions of the original sample (Reverse, Half-time, Double-time, Octave Up/Down).

2. INSTALLATION REQUIREMENTS

To run the script, you need to install:

1. **REAPER** (Latest version).
2. **RealmGui API:** Install via ReaPack (Search for "RealmGui").
3. **SWS Extension:** (Recommended) To support accurate RMS measurement.

3. INTERFACE & PARAMETERS

A. Top Bar

- **Detection Mode (Combo box):**
 - *RMS (Default):* Used for sustained sounds (Pads, Vocals, Guitars).
 - *Transient (Peak):* Used for fast, sharp sounds (Drums, Percussion).

- **Gear Icon (Settings):** Click to open advanced settings:
 - *Naming Scheme:* Naming convention (By Track, by Item, or simply Section 1, 2...).
 - *Smart Info:* Automatically append _120bpm or Key _Cm to filenames.
 - *Color Mode:* Colorize Regions by Track, Item, or Randomly.
 - *Normalize Target:* Set the target dB level for volume normalization.

B. Main Controls (Sliders)

- **Threshold (dB):** Volume threshold. Any sound louder than this level will be considered the start of a Loop.
 - *Tip:* Set lower (-40dB) for quiet sounds, higher (-10dB) for loud sounds.
- **Silence (s):** Minimum silence duration required to determine the "end" of a Loop.
 - *Example:* If set to 2.0s, the script must detect a full 2 seconds of silence before closing the old loop and searching for a new one.
- **Min Loop (s):** Minimum duration of a Loop. Shorter segments will be ignored (to avoid noise/artifacts).

C. Processing Options (Checkboxes)

- **Trim Excess:** Crop the excess head/tail of the original Item to match the Loop boundaries.
- **Snap to Grid:** Force Loop start/end points to snap to REAPER's Grid.
- **Auto Fades:** Add a 10ms fade at the start/end for smooth audio.
- **Split Items:** Physically split the Item at Marker points (Creates child items).
- **Strip Silence:** Delete items located in silent zones (Keep only the Loop).

4. DETAILED USER GUIDE

Step 1: Preparation

1. Import audio or MIDI files into REAPER.
2. **Important:** Select the Items you want to process on the Timeline using your mouse.

Step 2: Scan & Create Markers

1. Open the **Auto Loop Detector PRO** script.
2. Select a suitable **Preset** (e.g., "Drums" or "Vocals") or manually adjust the **Threshold** slider.
3. Press the large **SCAN** button at the bottom.
4. *Result:* The script will draw Regions and Markers on the Timeline surrounding the audio segments.
5. Use the navigation buttons (⏮ ▶ 🟡 ■ ▶▶) on the script interface to preview each found Loop.

Step 3: Map to Sampler (Create Playable Instrument)

If you want to play these slices on a MIDI keyboard:

1. After Scanning, ensure options like **Split Items** are enabled if you wish to separate the files.
2. Press the **Map to MIDI** button.
3. The script will create a new track named "Hosi Sampler".
4. Slices are loaded into **ReaSamplomatic5000**, mapped starting from key **C4 (Note 60)** upwards.
5. You can now use a MIDI Keyboard to play these samples.

Step 4: Create Variations (Sound Design)

1. Select the original Items.
2. Press the **Variations** button.
3. A new track named "Variations" will appear, containing versions:
 - Reverse.
 - Double Speed / Half Speed.
 - Pitch shift +/- 1 Octave.

Step 5: Export Sample Pack

This is a PRO feature for creating professional Sample Packs:

1. In the **Export Options** section (bottom of interface):
 - **Render Preset:** Select a render configuration (e.g., MP3, WAV 24bit...).
 - **Bounds:** "Project Regions" is recommended (to export each Loop as a single file).
 - **Format:** Select the desired format. This is the most powerful feature of the PRO version:
 - **DecentSampler (.dspreset):** Automatically creates a complete virtual instrument file for the free DecentSampler plugin. The script automatically designs the UI with knobs for: **Volume**, **Tone** (Lowpass Filter), and **Reverb**. Samples are auto-mapped starting from C3.
 - **SFZ Instrument (.sfz):** Open format compatible with many samplers like Sforzando, Zampler, etc. The script sets up Monophonic mode (prevents overlap) and a basic Envelope (short Release 0.1s).
 - **WAV Only:** Exports only raw audio files.
 - **All Formats:** Exports all the above formats simultaneously.
2. Press the **Export** button.
3. Enter a name for the instrument (e.g., "LoFi_Piano_Kit").
4. REAPER's **Render to File** dialog will appear with the path and filename already pre-configured.
5. Click **Render X files** in the REAPER dialog to finish.

The script will create a folder containing:

- Audio files (.wav).
- Instrument files (.dspreset, .sfz).
- An index.html file (Open this in a browser to view the web-based sample preview)

interface).

5. TIPS & TRICKS

- **Easter Egg:** Try Right-Clicking the "Variations" button to activate the hidden "Secret Un-Reverse" feature.
- **Auto-Velocity Map:** In Settings (Gear icon), if "Auto-Velocity Map" is enabled, when exporting, the script measures the loudness of each Loop and stacks them on a single key. Soft key press -> quiet sample, Hard key press -> loud sample.
- **Rescue (Undo):** All script actions are wrapped in a single Undo block. If the result isn't satisfactory, just press Ctrl+Z in REAPER to revert to the initial state.